002428.SZ - Yunnan Germanium / 云南锗业 | Full investment artifact

Full investment research | Model assumptions and source evidence

Public figures use whole-number presentation. Exact calculations, share counts, operating costs and every probability control remain in the full model JSON. Original source graphics retain their original units and precision. **Research date:** 19 October 2026 | **Framework:** 8.0 | **Format:** full current-state research **Status:** Complete new-source review and rescore; existing operating path retained. **Artifact analytical Base:** 39/60 = 65 | **Protected portfolio Excel Base:** 65. **Active ceiling:** Q4 2027, about CNY 26 per share, using Q4 2027 through Q3 2028. **Dated price context:** supplied-workbook snapshot about CNY 100; no current-market refresh is claimed.

The earnings inflection is supported by reported product margins. The long-term case must be rebuilt from disclosed wafer capacity and attributable earnings, without relying on unverified monopoly or six-inch volume claims.

October 19 current source review and full rescore

The complete new InP source was reviewed, the seven conviction categories were reassessed, and the full affected financial path was compared with the current master. All numerical operating inputs, valuation methods, shares and protected Excel Base values are retained.

Source identity, product stages and contrary views

P Equity Research published the 16-page InP Expert Call on October 19, 2026. The call is dated October 2026 without an exact day. The expert is described as China-based; name, employer and credentials are not supplied. The author's commentary and expert's views are separate evidence, rather than two independent interviews. No source-authored probability is provided. The original is retained with SHA-256 b457250e01dc31744a15836b260540bfdaf1348dcb9862e6cc2c096bf2b98ddc.

The expert expects capacity to meet demand around H2 2027, some 2027 price easing and clearer oversupply in 2028. The publisher disagrees and cites qualification and yield constraints that could prolong tightness through 2028. Both views remain visible. The expert's three-inch-equivalent totals and Figure 25's four-inch-equivalent chart have an unresolved axis multiplier; neither is converted into company sales. The claim that high-power chips require larger die area and lower defects describes resource intensity, with no universal supplier-volume multiplier.

Fewer than 10 of 100 grown crystals reportedly become finished six-inch wafers; much of the remainder is downgraded into saleable three- or four-inch products. This is a diameter/grade finishing ratio, not 90% material scrap, laser-die yield or module good-output yield. Customer qualification, export permission, substrate finishing, epitaxy, device processing and module acceptance remain distinct gates. Fixed-price and fixed-volume long-term contracts do not automatically reset after spot quotes change.

The interview describes roughly 5% recent price gains, diameter-specific domestic quotes and a 10-20% overseas contract premium. Grade, contract date, processing and customer terms differ. Its 18-CW-source CPO example belongs to a particular architecture. The raw-phosphorus and long-MOCVD-lead-time claims do not identify a named company allocation, delivery failure or incremental invoice. They are not mapped to every optical supplier.

Company profit capture, overlap and bounded proxy

Yunnan's compound-semiconductor vector combines InP and GaAs. The present company split is undisclosed. Existing capacity assumptions retain the staged path toward 450,000 four-inch-equivalent InP wafers, including the six-inch component within that total. The expert's generic 20,000-30,000 monthly physical wafers omit diameter and acceptance scope. They cannot be added to the company's existing annual equivalent capacity. Domestic three-inch spot quotes cannot be divided by the model's realized four-inch-equivalent revenue proxy to manufacture a company price gain.

The price diagnostic assigns 15% probability to a conditional 5% increase on 25% resettable InP receipts. Its explicitly assumed InP share is 50%, with a 25-75% range; that split is not a current issuer disclosure. Extra manufacturing/support costs consume 5% of price revenue, the existing quarter opex ratio is charged, and the existing economic attribution is applied once. The midpoint adds about CNY 795,000 to four-quarter parent operating profit before overlap, equivalent to about 3 fen per share. The diagnostic gross revenue uplift is about 9 basis points of the combined vector.

The selected increment remains zero. H1 compound gross margin of about 47% already rises to 49% in Q3 2026 and remains about 48% in Q4 2027 before normalization. No comparable invoice identifies a price cohort above that whole operating path. Allocating unverified quotes across all InP/GaAs revenue would overstate both scope and capture. The full assessment retains a 65 analytical score and protected Excel Base 65. The active Q4 2027 ceiling is about CNY 26, with no economic change. The old Q3 2027 headline and old Excel 72 display are corrected to the existing master conventions; those display corrections create no model profit.

Issuer cross-check and required evidence

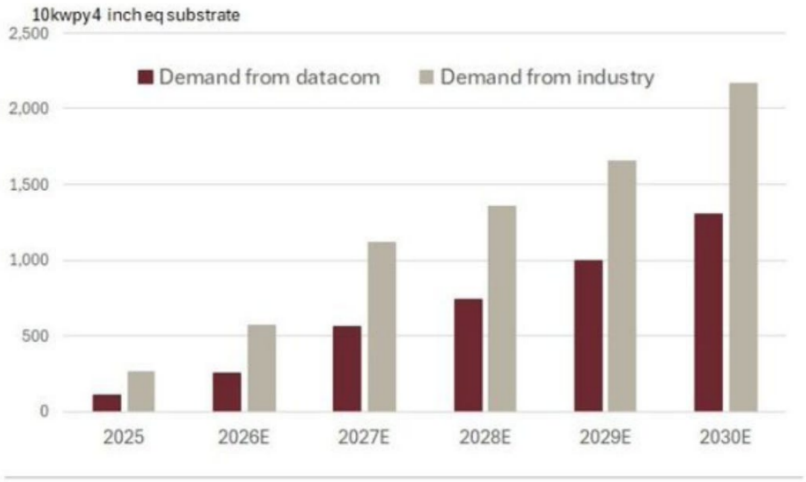
The September 11 issuer record describes existing physical two- to four-inch capacity, the 300,000 four-inch-equivalent expansion and its 18-month construction period. Its six-inch physical quantity is part of the expansion rather than a second capacity pool. It reports normal domestic raw-material procurement and equipment delivery. The anonymous upstream supply concern does not establish a missed Yunnan input or machine delivery.

Counter-evidence remains meaningful: extra competing capacity, customer inventory, price easing, qualification delays, germanium substitution and the contingent repurchase liability can reduce economic capture. Next evidence is the InP/GaAs revenue and margin split; accepted good output by diameter; realized contract reset dates and prices; ownership versus accounting attribution; and cash required for expansion/repurchase obligations. The existing later margin decline already recognizes normalization rather than permanent scarcity.

September 11 issuer record, read as a dated management summary. General expert claims do not replace those product-specific disclosures.

Original source pages

Figure 25: InP substrates demand forecast



Source: Yole,Company data, UBSe

Q6. What is your view of the capacity expansion plans of major

P Equity Research InP Expert Call, published October 19, 2026, original page 9. Expert channel checks and publisher interpretation remain attributed; original units are preserved. Original exhibit, panel 1 of 2; read from top to bottom.

manufacturers such as AXT and Yunnan Germanium, and their effect on market supply next year (2027)?

The capacity targets of 200,000 or 300,000 wafers mentioned by AXT and Yunnan Germanium should be long-term plans. By 2027, the limit of their capacity expansion is roughly double 2026 levels. Even so, AXT's monthly capacity could then reach nearly 100,000 wafers, with annual capacity above 1 million. Yunnan Germanium and Xiandao will not be small either; together the three are expected to achieve annual capacity of more than 3 million wafers in 2027. Therefore, by the second half of 2027, the tight market supply is expected to ease.

Q7. How have InP substrate prices changed recently, and how should future price trends be judged?

Prices have risen slightly recently, by about 5%, but far less than the rise in the first half of 2026. Specifically, 2-inch products were around RMB 1,500 last month (September 2026); 3-inch products are close to RMB

P Equity Research InP Expert Call, published October 19, 2026, original page 9. Expert channel checks and publisher interpretation remain attributed; original units are preserved. Original exhibit, panel 2 of 2; read from top to bottom.

has built a 30-ton line, with current monthly output of about 300–500 kg, and capacity is still ramping. In addition, Guizhou Jinding and Wuhan Tuocai also have small outputs. Because Japanese suppliers are tight, and a UK InP polycrystal supplier has halted production for being unable to buy electronic-grade phosphorus, some domestic high-purity phosphorus products have entered InP polycrystal supply chains for validation.

Q14. On the equipment side, are there supply bottlenecks in crystal growth furnaces and MOCVD equipment?

There is no supply bottleneck in crystal growth furnaces. VGF equipment is relatively simple in structure; the core lies in each manufacturer's process parameter tuning rather than the equipment itself. Delivery lead time for domestic furnaces is only one to two months. By comparison, the VB process demands more of equipment. The main problem facing the industry now is raw material supply cutoff, not furnace equipment. In the downstream epitaxy step, **MOCVD**

P Equity Research InP Expert Call, published October 19, 2026, original page 13. Expert channel checks and publisher interpretation remain attributed; original units are preserved. Original exhibit, panel 1 of 2; read from top to bottom.

equipment lead times are indeed long, mainly because Aixtron's capacity cannot meet market demand.

Q15. What is the latest progress in InP substrate cutting technology, and how is diamond wire cutting being adopted?

The industry is currently transitioning from traditional slurry wire cutting to diamond wire cutting. In the past, InP cutting was mainly dominated by slurry wire cutting equipment from two Japanese companies, Yasunaga and Takatori. Now the industry is generally introducing diamond wire cutting, but it is still at an early stage. Many new entrants usually first buy one or two slurry wire cutters to ensure normal production and product shipment, and also buy one diamond wire cutter for process tuning and development. Although slurry wire cutting is still the mainstay, the shift to diamond wire has become an industry consensus, because diamond wire is thinner and can cut more wafers from the same ingot, raising output. By 2027, diamond wire adoption is

P Equity Research InP Expert Call, published October 19, 2026, original page 13. Expert channel checks and publisher interpretation remain attributed; original units are preserved. Original exhibit, panel 2 of 2; read from top to bottom.

1. Supercycle, supply chain and durable advantage

AI optical links need qualified compound-semiconductor substrates, while space power and fiber systems consume germanium-based materials. Yunnan Germanium has established production and customer validation, but substrates, epitaxy, laser devices and modules are separate markets. Demand growth expands its opportunity without proving exclusivity, guaranteeing higher InP content per module, or turning customer certification into a purchase commitment.

Raw germanium/indium/phosphorus → purified material/crystal → qualified substrate → epitaxy → device → optical module or solar cell. Module growth is several steps downstream; customer yields, wafer size and die design affect substrate demand. Revenue does not equal all module-market value.

Business / curve	Stage	Evidence and next inflection
InP	Commercial supply expanding	Existing customer qualifications; new line in construction.
GaAs	Relocation and upgrade	Six-inch line requires utilization and competitive capture.
Space solar	Capacity expansion / demand conversion	Capacity alone does not ensure orders.
Infrared	Substitution pressure	Higher germanium costs encourage alternative materials.

Stage percentages are qualitative framework bands relative to a possible mature opportunity, not measured market penetration. A current capacity constraint can coexist with a larger long-term TAM; qualification and economic capture decide how much reaches this company.

2. Forward TAM and the three-phase expansion lens

Phase	Engineering opportunity	Demand-to-company bridge
2026-27	AI optical materials	2027 compound760 million from staged InP expansion plus GaAs upgrade; no million-wafer InP assumption.
2028-29	Higher qualified wafer utilization	2029 compound1,560 million requires demand and improving product mix; build only within disclosed capacity paths.
2030+	Optical and space opportunity	Compound1,740 million bridge:450k four-inch-equivalent InP capacity×90% utilization×2200CNY realized revenue/equivalent=891 million;700k six-inch GaAs×60%×2,021CNY≈849 million. Solar920 million=2,500,000 capacity×70%×526CNY. These ASPs/utilizations are analyst assumptions.

The 100% / 50% / 20% thematic weights guide thesis assessment. They are not blanket discounts to reported sales, contracts or all forecasts. Programme probabilities below govern only separately identified optionality. Current reported sales calibrate the starting point; they do not cap the future market.

3. Current earnings and management evidence

H1 revenue732 million, GP204 million and core OP100 million after operating taxes, selling, administration and expensed R&D [E1]. Parent net income74 million is a different measure. Compound revenue142 million has47% reported GM, showing a real improvement in economics; legacy categories remain mixed.

The September4 and September11 published management Q&A were read in full [E2,E3]. They confirm mostly3-4inch InP demand, staged capacity release and no disclosed further expansion plan. Current6inch unit volumes, numeric yield percentages and individual customer contracts were not supplied. Management discusses cost pass-through negotiations and substitution pressure in infrared products.

Two latest issuer Q&A records were reviewed completely; these are published summaries rather than verbatim earnings-call audio. InP/GaAs revenue split, actual customer contract quantities and exact numeric yields remain undisclosed.

4. Local evidence and calibration packet

H1 production91,800 InP wafers is a physical2-4inch count; it cannot be directly divided by a future four-inch-equivalent capacity without a size conversion. Calibrate to reported product revenue and GP with0% total error. H1 core cost bridge is5,543,115 taxes+10,239,345 selling+41,306,782 administration+46,594,074 R&D. No disclosed customer capex allocation is available; AH12 uses capacity×utilization×realized ASP, with units kept consistent.

5. Summing thesis score

Category	Points / maximum	Current review reason
Core thesis	13/20	Qualified substrate production and domestic resources retain value, but no new exclusive customer or proven relative cost advantage emerges.
Cycle position	8/10	H1 compound improvement and new capacity remain favorable. Expert price easing and inventory demand are valid counters already consistent with caution.
Forward TAM	9/10	Optical, fiber and space-power vectors remain. Larger high-power die can support InP use; capacity and product split constrain capture.
Catalysts	4/5	The dated18-month expansion and GaAs relocation remain; source furnaces and generic monthly output do not establish a new company acceptance date.
Macro and policy	7/10	Domestic optics demand supports volume while licensing and substitution constrain scope.
Alpha and evidence	3/5	Correct physical-versus-equivalent wafer units, InP/GaAs split and parent attribution remain the edge. New anonymous prices do not resolve realized company ASP.
Risk deduction	-5	Retain capacity/qualification; competition/price; materials/substitution; cash/capital structure; export/ownership.

Analytical Base 39/60 = 65; protected Excel Base 65. The score is independent of share price and does not multiply valuation.

6. Revenue sources, capture and product margins

Six reported product categories replace an unsupported six-inch volume-led model. Historical anchors sum exactly to H1 revenue. Each H1 margin is computed from disclosed sales and costs: compound47%, materials16%, solar29%, fiber26%, infrared44%, other56% approximately. Forward product margins are distinct and explicitly assumed. InP and GaAs remain one vector because their revenue split is not reported; a capacity bridge below limits the2030 combined estimate.

Vector	Actual calibration / source	Forward demand and capture	Margin derivation / constraint
Compound semiconductor substrates	H1 revenue142 million reported	Forward factor is quarterly demand/utilization relative to H1 anchor; see capacity bridge and quarterly tables	Separate product GM path; all future assumptions LOW
Materials-grade germanium	H1 revenue330 million reported	Forward factor is quarterly demand/utilization relative to H1 anchor; see capacity bridge and quarterly tables	Separate product GM path; all future assumptions LOW
Space solar germanium wafers	H1 revenue90 million reported	Forward factor is quarterly demand/utilization relative to H1 anchor; see capacity bridge and quarterly tables	Separate product GM path; all future assumptions LOW
Fiber-grade germanium	H1 revenue107 million reported	Forward factor is quarterly demand/utilization relative to H1 anchor; see capacity bridge and quarterly tables	Separate product GM path; all future assumptions LOW
Infrared germanium optics	H1 revenue34 million reported	Forward factor is quarterly demand/utilization relative to H1 anchor; see capacity bridge and quarterly tables	Separate product GM path; all future assumptions LOW
Other materials and activities	H1 revenue29 million reported	Forward factor is quarterly demand/utilization relative to H1 anchor; see capacity bridge and quarterly tables	Separate product GM path; all future assumptions LOW

Confidence: reported historical totals are HIGH. Source-based forward extrapolations are MEDIUM only where the cited source sizes the relevant period; other product allocations, future capture and margin assumptions are LOW. A source for the historical anchor does not make an unsourced future factor management guidance. The detailed factors below are the single modeled base case, with uncertain assumptions exposed for review.

V1. Compound semiconductor substrates

Sources: E1/E2/E3; analyst utilization/ASP. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor, CNY M	Conversion versus anchor	Revenue, CNY M	Product margin	GP, CNY M
Q3 2026	142	70%	100	49%	49
Q4 2026	142	91%	130	49%	63
Q1 2027	142	99%	140	49%	68
Q2 2027	142	120%	170	48%	82
Q3 2027	142	141%	200	48%	97
Q4 2027	142	176%	250	48%	120
Q1 2028	142	162%	230	48%	110
Q2 2028	142	190%	270	48%	129
Q3 2028	142	211%	300	48%	143
Q4 2028	142	246%	350	47%	166
Q1 2029	142	232%	330	47%	156
Q2 2029	142	260%	370	47%	174
Q3 2029	142	289%	410	47%	192
Q4 2029	142	317%	450	47%	210
Q1 2030	142	274%	390	47%	181
Q2 2030	142	296%	420	46%	195
Q3 2030	142	317%	450	46%	208
Q4 2030	142	338%	480	46%	221

V2. Materials-grade germanium

Sources: E1/E2; analyst volume and pricing. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor, CNY M	Conversion versus anchor	Revenue, CNY M	Product margin	GP, CNY M
Q3 2026	330	55%	180	18%	32
Q4 2026	330	58%	190	18%	34
Q1 2027	330	55%	180	18%	33
Q2 2027	330	61%	200	18%	37
Q3 2027	330	64%	210	18%	39
Q4 2027	330	67%	220	19%	41
Q1 2028	330	61%	200	19%	37
Q2 2028	330	67%	220	19%	41
Q3 2028	330	70%	230	19%	44
Q4 2028	330	76%	250	19%	48
Q1 2029	330	67%	220	19%	42
Q2 2029	330	73%	240	19%	46
Q3 2029	330	76%	250	19%	49
Q4 2029	330	82%	270	20%	53
Q1 2030	330	70%	230	20%	45
Q2 2030	330	76%	250	20%	49
Q3 2030	330	79%	260	20%	52
Q4 2030	330	85%	280	20%	56

V3. Space solar germanium wafers

Sources: E1/E2/E3; analyst satellite utilization. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor, CNY M	Conversion versus anchor	Revenue, CNY M	Product margin	GP, CNY M
Q3 2026	90	66%	60	29%	17

Quarter	Revenue anchor, CNY M	Conversion versus anchor	Revenue, CNY M	Product margin	GP, CNY M
Q4 2026	90	89%	80	29%	23
Q1 2027	90	89%	80	30%	24
Q2 2027	90	111%	100	30%	30
Q3 2027	90	133%	120	30%	36
Q4 2027	90	155%	140	30%	43
Q1 2028	90	133%	120	31%	37
Q2 2028	90	166%	150	31%	47
Q3 2028	90	199%	180	31%	56
Q4 2028	90	232%	210	32%	66
Q1 2029	90	188%	170	32%	54
Q2 2029	90	221%	200	32%	64
Q3 2029	90	255%	230	33%	75
Q4 2029	90	288%	260	33%	85
Q1 2030	90	221%	200	33%	66
Q2 2030	90	243%	220	33%	74
Q3 2030	90	266%	240	34%	81
Q4 2030	90	288%	260	34%	88

V4. Fiber-grade germanium

Sources: E1/E2; analyst cost pass-through. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor, CNY M	Conversion versus anchor	Revenue, CNY M	Product margin	GP, CNY M
Q3 2026	107	56%	60	27%	16
Q4 2026	107	66%	70	27%	19

Quarter	Revenue anchor, CNY M	Conversion versus anchor	Revenue, CNY M	Product margin	GP, CNY M
Q1 2027	107	66%	70	27%	19
Q2 2027	107	75%	80	28%	22
Q3 2027	107	84%	90	28%	25
Q4 2027	107	94%	100	28%	28
Q1 2028	107	84%	90	28%	25
Q2 2028	107	94%	100	28%	28
Q3 2028	107	103%	110	28%	31
Q4 2028	107	112%	120	29%	34
Q1 2029	107	103%	110	29%	32
Q2 2029	107	112%	120	29%	35
Q3 2029	107	122%	130	29%	38
Q4 2029	107	131%	140	29%	41
Q1 2030	107	112%	120	29%	35
Q2 2030	107	122%	130	30%	39
Q3 2030	107	131%	140	30%	42
Q4 2030	107	141%	150	30%	45

V5. Infrared germanium optics

Sources: E1/E2; analyst substitution. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor, CNY M	Conversion versus anchor	Revenue, CNY M	Product margin	GP, CNY M
Q3 2026	34	54%	18	42%	8
Q4 2026	34	59%	20	42%	8
Q1 2027	34	59%	20	42%	8

Quarter	Revenue anchor, CNY M	Conversion versus anchor	Revenue, CNY M	Product margin	GP, CNY M
Q2 2027	34	68%	23	41%	9
Q3 2027	34	74%	25	41%	10
Q4 2027	34	80%	27	41%	11
Q1 2028	34	68%	23	41%	9
Q2 2028	34	74%	25	40%	10
Q3 2028	34	80%	27	40%	11
Q4 2028	34	89%	30	40%	12
Q1 2029	34	77%	26	40%	10
Q2 2029	34	83%	28	39%	11
Q3 2029	34	89%	30	39%	12
Q4 2029	34	98%	33	39%	13
Q1 2030	34	83%	28	39%	11
Q2 2030	34	89%	30	38%	12
Q3 2030	34	95%	32	38%	12
Q4 2030	34	104%	35	38%	13

V6. Other materials and activities

Sources: E1; analyst residual operations. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor, CNY M	Conversion versus anchor	Revenue, CNY M	Product margin	GP, CNY M
Q3 2026	29	51%	15	40%	6
Q4 2026	29	55%	16	40%	6
Q1 2027	29	55%	16	40%	6
Q2 2027	29	58%	17	40%	7

Quarter	Revenue anchor, CNY M	Conversion versus anchor	Revenue, CNY M	Product margin	GP, CNY M
Q3 2027	29	61%	18	40%	7
Q4 2027	29	65%	19	40%	8
Q1 2028	29	58%	17	40%	7
Q2 2028	29	61%	18	40%	7
Q3 2028	29	65%	19	40%	8
Q4 2028	29	68%	20	40%	8
Q1 2029	29	61%	18	40%	7
Q2 2029	29	65%	19	40%	8
Q3 2029	29	68%	20	40%	8
Q4 2029	29	72%	21	40%	8
Q1 2030	29	65%	19	40%	8
Q2 2030	29	68%	20	40%	8
Q3 2030	29	72%	21	40%	8
Q4 2030	29	75%	22	40%	9

7. Operating-cost bridge and quarterly operating model

Explicit quarterly cost60→105M includes operating taxes, R&D, selling and administration. Depreciation remains in product costs/expenses. Of these costs, semiconductor20→45M, solar10→20M and fiber14→27M are allocation estimates used solely to calculate minority shares; they are already included in total opex and not deducted twice. Capitalized R&D and working capital remain cash-flow limitations.

Profit basis: Product GP less operating taxes, selling, administration and expensed R&D, before finance, investment gains and operating subsidies; minority allocation included in ceilings. Current operating tables below use CNY millions; the full model retains its exact CNY-billion base units.

Quarter	Status	Revenue, CNY M	GP / contribution, CNY M	Opex %	Opex, CNY M	OP margin	Operating profit, CNY M	Main driver	Confidence
Q3 2026	Model	433	129	14%	60	16%	69	InP pricing and initial installation	LOW model

Quarter	Status	Revenue, CNY M	GP / contribution, CNY M	Opex %	Opex, CNY M	OP margin	Operating profit, CNY M	Main driver	Confidence
Q4 2026	Model	506	155	12%	63	18%	92	Customer conversion	LOW model
Q1 2027	Model	506	158	13%	65	18%	93	GaAs relocation progress	LOW model
Q2 2027	Model	590	187	12%	68	20%	119	Higher qualified production	LOW model
Q3 2027	Model	663	214	11%	71	22%	143	InP expansion near planned completion	LOW model
Q4 2027	Model	756	250	10%	73	23%	177	Full-line utilization begins	LOW model
Q1 2028	Model	680	226	11%	76	22%	150	New product qualification	LOW model
Q2 2028	Model	783	262	10%	79	23%	184	Optical demand	LOW model
Q3 2028	Model	866	292	9%	81	24%	211	Solar capacity uptake	LOW model
Q4 2028	Model	980	334	9%	84	26%	250	Customer deliveries	LOW model
Q1 2029	Model	874	302	10%	86	25%	215	Yield versus price normalization	LOW model
Q2 2029	Model	977	338	9%	89	26%	249	GaAs utilization	LOW model
Q3 2029	Model	1,070	373	9%	92	26%	281	Serviceable materials mix	LOW model
Q4 2029	Model	1,174	411	8%	94	27%	316	Space solar demand	LOW model
Q1 2030	Model	987	347	10%	97	25%	250	2030 utilization case	LOW model

Quarter	Status	Revenue, CNY M	GP / contribution, CNY M	Opex %	Opex, CNY M	OP margin	Operating profit, CNY M	Main driver	Confidence
Q2 2030	Model	1,070	376	9%	100	26%	276	Compound product mix	LOW model
Q3 2030	Model	1,143	403	9%	102	26%	300	Pricing and competitive response	LOW model
Q4 2030	Model	1,227	432	9%	105	27%	327	Capacity-boun d endpoint	LOW model

OP = sum of product GP/contribution - explicit operating costs. OP margin = OP ÷ revenue. Displayed numbers are rounded for reading; the workbook calculates from unrounded inputs. Guidance assigned to a future quarter retains Model status because guidance is not an actual result.

8. Multiple, shares and recognition convention

Retain25× four-quarter operating profit as the established specialty-materials convention. No monopoly premium or unsupported historical peer P/E claim is needed. Finance costs and taxes are outside this OP convention and must remain visible as economic limitations.

653,120,000 shares per H1 report, held constant. No new equity issuance is invented; financing and incentive risk remains in the score and dilution sensitivity. Net-debt and repurchase liabilities are not silently removed from the risk analysis simply because the framework uses an OP ceiling.

The valuation denominator is the ceiling quarter’s diluted-share proxy. It is not a historical EPS weighted-average assumption disguised as a fully known future capitalization. Unit factor: 1000; this converts the model units to CNY per share. No cash balance, enterprise-value adjustment or after-tax earnings substitution is added to the operating-profit ceiling.

Minority earnings allocation

H1 accounting table attributes95% of Xinyao, explicitly including interests subject to repurchase obligations, with5% minority. Use5% of estimated compound-plus-solar operating profit and40% of fiber operating profit for the minority deduction. This is an accounting-economic proxy, not a claim of95% registered ownership. The462,392,529 contingent repurchase liability is a material funding risk; other small subsidiaries are assumed immaterial.

Quarter	Consolidated operating profit, CNY M	Minority allocation proxy, CNY M	Parent economic OP proxy, CNY M
Q3 2026	69	3	66
Q4 2026	92	4	88
Q1 2027	93	4	89
Q2 2027	119	6	113

Quarter	Consolidated operating profit, CNY M	Minority allocation proxy, CNY M	Parent economic OP proxy, CNY M
Q3 2027	143	8	136
Q4 2027	177	10	167
Q1 2028	150	8	142
Q2 2028	184	10	174
Q3 2028	211	12	199
Q4 2028	250	15	236
Q1 2029	215	12	203
Q2 2029	249	14	235
Q3 2029	281	16	265
Q4 2029	316	19	297
Q1 2030	250	14	236
Q2 2030	276	16	260
Q3 2030	300	17	283
Q4 2030	327	19	308

9. Complete rolling four-quarter ceiling derivation

Ceiling quarter	Exact earnings window	F4Q parent OP, CNY M	Multiple	Diluted shares	Ceiling CNY	Key inflection
Q3 2026	Q3 2026 → Q2 2027	356	25×	653,120,000	14	InP pricing and initial installation
Q4 2026	Q4 2026 → Q3 2027	426	25×	653,120,000	16	Customer conversion
Q1 2027	Q1 2027 → Q4 2027	505	25×	653,120,000	19	GaAs relocation progress
Q2 2027	Q2 2027 → Q1 2028	558	25×	653,120,000	21	Higher qualified production
Q3 2027	Q3 2027 → Q2 2028	619	25×	653,120,000	24	InP expansion near planned completion
Q4 2027	Q4 2027 → Q3 2028	682	25×	653,120,000	26	Full-line utilization begins
Q1 2028	Q1 2028 → Q4 2028	751	25×	653,120,000	29	New product qualification
Q2 2028	Q2 2028 → Q1 2029	812	25×	653,120,000	31	Optical demand
Q3 2028	Q3 2028 → Q2 2029	873	25×	653,120,000	33	Solar capacity uptake
Q4 2028	Q4 2028 → Q3 2029	939	25×	653,120,000	36	Customer deliveries
Q1 2029	Q1 2029 → Q4 2029	1,000	25×	653,120,000	38	Yield versus price normalization
Q2 2029	Q2 2029 → Q1 2030	1,033	25×	653,120,000	40	GaAs utilization
Q3 2029	Q3 2029 → Q2 2030	1,059	25×	653,120,000	41	Serviceable materials mix
Q4 2029	Q4 2029 → Q3 2030	1,077	25×	653,120,000	41	Space solar demand
Q1 2030	Q1 2030 → Q4 2030	1,087	25×	653,120,000	42	2030 utilization case

10. Annual trajectory cross-check

Calendar year	Revenue, CNY M	GP / contribution, CNY M	Opex, CNY M	OP, CNY M	OP margin
2027	2,515	810	277	533	21%
2028	3,309	1,115	319	796	24%
2029	4,095	1,423	362	1,062	26%
2030	4,427	1,557	404	1,153	26%

These are calendar-year sums of the quarterly path, not extra revenue and not the live valuation method. For non-December fiscal reporters, fiscal guidance and these calendar totals must be aligned using the period mapping above; a four-quarter run rate is not automatically the same fiscal year. Outer-year outputs are conditional TAM/capture trajectories, not verified management sales targets.

2030 compound1,740 million+materials1,020 million+solar920 million+fiber540 million+infrared125 million+other82 million=4,427 million. Larger market demand is not automatically captured above disclosed manufacturing capacity. The model can expand if a new funded project is announced; none is presumed completed.

11. Dilution sensitivity - same operating case

Denominator sensitivity	Q4 2027 shares M	Q4 2027 ceiling CNY	Explanation
-5%	620	27	Same profit; denominator only
0%	653	26	Same profit; denominator only
5%	686	25	Same profit; denominator only
10%	718	24	Same profit; denominator only

No new financing is selected; contingent liabilities and future equity remain specific risks.

12. Programme and probability ledger

Programme / signal	Source and probability	Amount / unit base	Treatment in this model	Escalation / recognition
InP expansion	E3;75% timely utilization assumption	450k four-inch-equivalent terminal capacity	Quarterly ramp; no duplicate six-inch overlay	Installed tools, qualification and sales
Space solar	E2/E3;70%2030 utilization	2,500,000 wafers capacity	Embedded in solar revenue	Actual purchases and cost pass-through
Unverified Huawei order	No reliable contract evidence;0% modeled	Unquantified	No separate revenue	Require dated issuer/customer disclosure

Probabilities are analyst judgments under the framework signal hierarchy. Embedded revenue is not multiplied or added a second time. Where no defensible amount or unit base exists, the opportunity remains explicitly unquantified; confidence alone cannot create a revenue number. New evidence can enlarge the TAM, improve capture or move recognition earlier, but each requires an identifiable model input.

13. Scored risk classes

Risk class	Scored deduction	Mechanism and consequence
Capacity and qualification	-1	New tools must produce customer-qualified saleable wafers.
Competition and price	-1	More substrate capacity can normalize scarcity margins.
Raw materials and substitution	-1	Germanium costs can suppress downstream demand.
Cash and capital structure	-1	Negative operating cash flow, debt and repurchase obligations require funding.
Export licensing and ownership	-1	Permits constrain shipments; accounting interests differ from registered equity.

14. Dated catalysts, counters and escalation

1. **Late2026 - Reported mix** Verify compound growth, collections and raw-material costs.
2. **2027 - Capacity qualification** Monitor18-month InP project fromApril2026 and GaAs relocation.
3. **Ongoing - Policy and ownership** Check actual permit rules and repurchase terms rather than assuming a single expiry date.

15. Cross-book monitoring

Compare substrate availability with AXT, IQE, LITE, COHR and optical-module suppliers. More qualified substrates can enable downstream volume while reducing scarcity margins upstream. No customer identity or sourcing exclusivity is inferred merely from an industry relationship.

16. Source register and model limits

E1 - Issuer H1 financial report (2026-08-28). Product revenue/costs, R&D, financing, production and subsidiary accounting.

E2 - Management investor Q&A (2026-09-04). Demand sizes, qualification, substitution and project plans.

E3 - Management investor Q&A (2026-09-11). Expansion450000 four-inch-equivalent InP wafers including6000 six-inch physical wafers in the added line; equipment installation progressing.

E4 - MOFCOM indium and related-material export controls (2025-02-04). InP is a listed export-controlled item; a licensing regime is not a ban on foreign supply to Chinese customers.

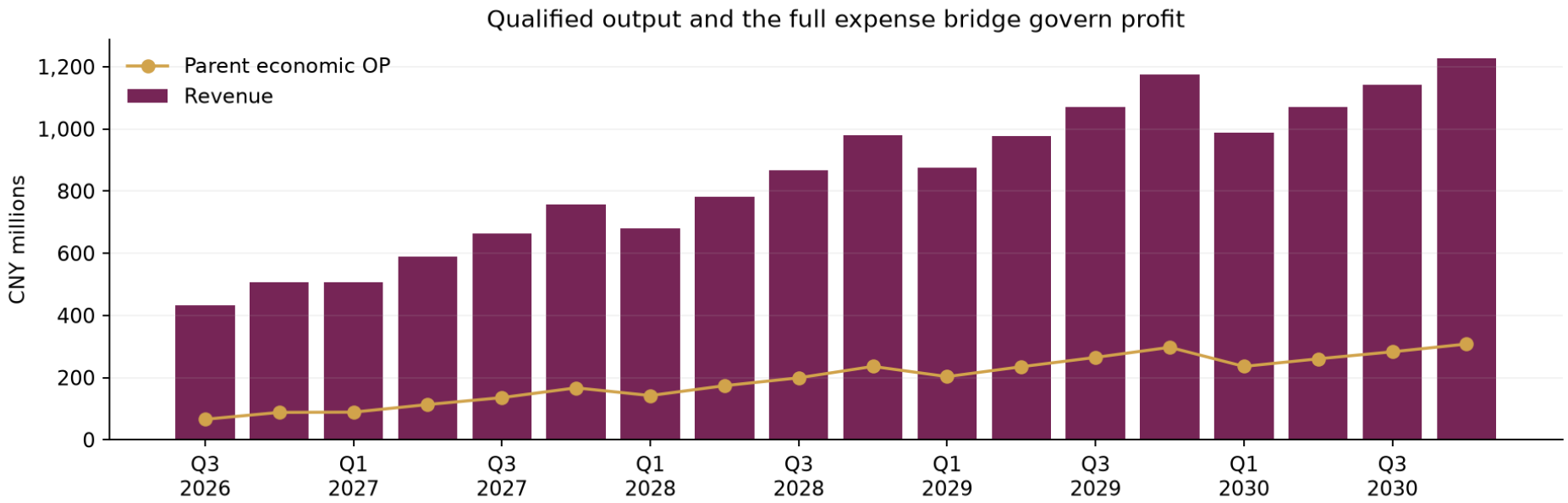
E5 - MOFCOM statement on suspended October2025 measures (2026-04). The statedNovember10 suspension date concerns specifiedOctober2025 measures and does not establish automatic expiry of the separateFebruary InP controls.

E6 - Xinyao capital increase disclosure (2024-12-20). Historical Hubble24% ownership is stale after subsequent capital increases; use current financial accounting rather than the old percentage.

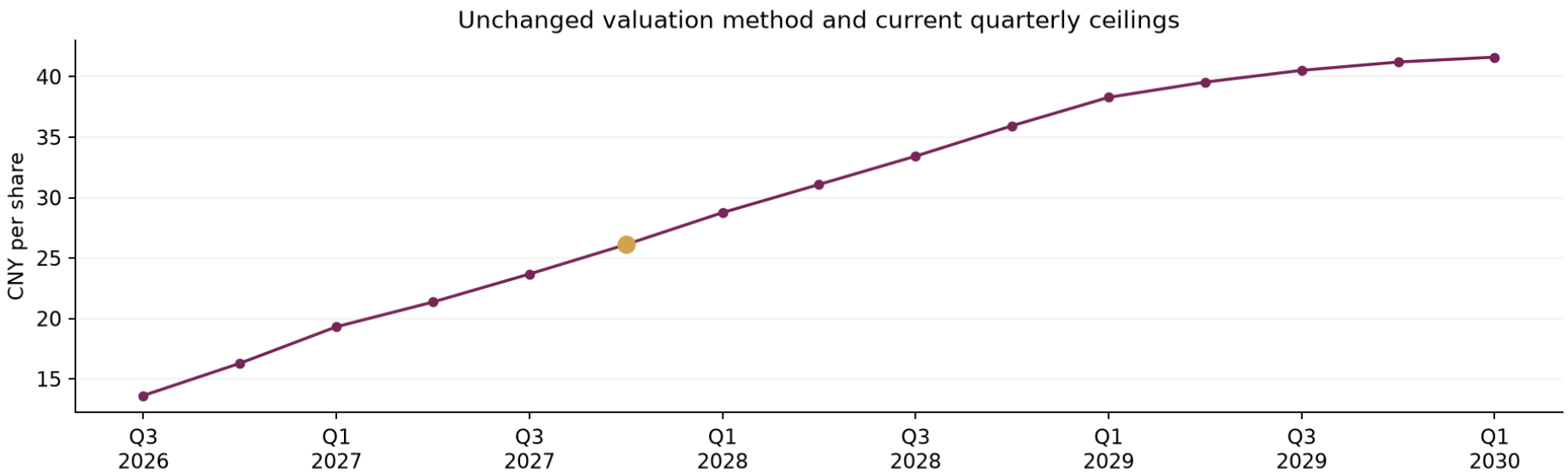
No verified80% domestic share, exclusive Huawei procurement,150,000 six-inch contracted units,60% six-inch production mix,1000000-wafer2027 project or guaranteed November10 InP-control expiry is assumed. This is a capacity-constrained operating scenario, not a company multi-year guide. Development capitalization24 million in H1 means operating profit is not cash generation.

Current workbook: v3_2_master_portfolio.xlsx, model tab 002428_SZ_Q_Model. Current local full model sidecar preserves all product and operating rows. Ranking and live prices are outside this unchanged financial review. **Current artifact:** 002428SZ_YunnanGermanium_DeepDive.md in the production Artifacts folder. **Original snapshot:** versions/originals/002428SZ_YunnanGermanium_DeepDive.md. Superseded history is retained separately in VERSION_LOG, not in this current-state model.

Current full model visual checks



Current revenue and full operating-profit path; financial inputs are unchanged.



Current ceiling path. Gold marks Q4 2027; methods, earnings windows and share conventions are retained.